Econometrics II: Panel Data
Graded Problem Set April 2026 - Cecile Macaire

Code to be found on my github: Github code

1 Exercise 1: First Differences vs Within Groups

We consider the static panel model

$$y_{it} = \beta x_{it} + \eta_i + v_{it}, \qquad i = 1, \dots, N, \quad t = 1, \dots, T,$$

where T is fixed and $N \to \infty$. We have strict exogeneity with respect to the instruments:

$$\mathbb{E}\left[v_{it} \mid \eta_i, z_{i1}, \dots, z_{iT}\right] = 0.$$

The individual effect η_i may be arbitrarily correlated with the instruments:

$$\mathbb{E}\left[\eta_i \mid z_{i1}, \dots, z_{iT}\right]$$

is left unrestricted. Hence the fixed-effect must be eliminated by a transformation.

Let

$$y_i = (y_{i1}, \dots, y_{iT})', \quad x_i = (x_{i1}, \dots, x_{iT})', \quad z_i = (z_{i1}, \dots, z_{iT})', \quad v_i = (v_{i1}, \dots, v_{iT})'.$$

The covariance matrix of v_i conditional on z_i is

$$\Sigma(z_i) = \mathbb{V}\mathrm{ar}(v_i \mid z_i).$$

The problem parametrizes this covariance structure using

$$v_{it} = \rho v_{i,t-1} + \varepsilon_{it}, \qquad t = 2, \dots, T,$$

with

$$\mathbb{V}\mathrm{ar}(v_{i1} \mid z_i) = \sigma^2, \qquad \mathbb{V}\mathrm{ar}(\varepsilon_{it} \mid z_i) = \sigma^2.$$

This is not the usual stationary AR(1) normalization, because the initial variance is set equal to σ^2, not $\sigma^2/(1-\rho^2)$. Therefore, the conditional covariance matrix is implied recursively.

Preliminaries: transformation matrices

It is useful to define two matrices.

First-difference matrix. Let D be the $(T-1)\times T$ first-difference matrix:

$$D = \begin{pmatrix} -1 & 1 & 0 & \cdots & 0 \\ 0 & -1 & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -1 & 1 \end{pmatrix}.$$

Then

$$\Delta y_i = Dy_i, \quad \Delta x_i = Dx_i, \quad \Delta z_i = Dz_i, \quad \Delta v_i = Dv_i.$$

Because $D\iota_T = 0$, where ι_T is a T-vector of ones, first differencing removes the fixed effect:

$$D(\eta_i \iota_T) = 0.$$

Within transformation matrix. Let

$$M_T = I_T - \frac{1}{T}\iota_T \iota_T'.$$

Then

$$\widetilde{y}_i = M_T y_i = y_i - \bar{y}_i \iota_T, \quad \widetilde{x}_i = M_T x_i, \quad \widetilde{z}_i = M_T z_i, \quad \widetilde{v}_i = M_T v_i.$$

Again $M_T \iota_T = 0$, so

$$M_T(\eta_i \iota_T) = 0.$$

1.1 Part (a): FD and WG 2SLS estimators and their asymptotic distributions

First-difference IV estimator

The first-difference equation is

$$\Delta y_i = \beta \Delta x_i + \Delta v_i.$$

The instruments are $\Delta z_i = Dz_i$. Since z_{it} is r-dimensional, Δz_i is a $(T-1)\times r$ stacked object if one thinks of one r-vector instrument per differenced equation. Equivalently, we can write the moment condition as

$$\mathbb{E}\left[\sum_{t=2}^{T} \Delta z_{it}\left(\Delta y_{it} - \beta \Delta x_{it}\right)\right] = 0.$$

The sample analogue is

$$\frac{1}{N}\sum_{i=1}^{N}\sum_{t=2}^{T}\Delta z_{it}\left(\Delta y_{it}-b\Delta x_{it}\right)=0.$$

Because β is scalar, the IV/2SLS estimator can be written compactly as

$$\widehat{\beta}_{FD}=\left[\left(\sum_i \Delta x_i' P_{\Delta Z_i}\Delta x_i\right)\right]^{-1}\left[\sum_i \Delta x_i' P_{\Delta Z_i}\Delta y_i\right],$$

where $P_{\Delta Z_i}$ is the projection matrix on the column space of the instruments. More generally, using all cross-sectional observations together, define

$$\Delta Y=\begin{pmatrix}\Delta y_1\\ \vdots\\ \Delta y_N\end{pmatrix},\quad \Delta X=\begin{pmatrix}\Delta x_1\\ \vdots\\ \Delta x_N\end{pmatrix},\quad \Delta Z=\begin{pmatrix}\Delta z_1\\ \vdots\\ \Delta z_N\end{pmatrix}.$$

Then

$$\widehat{\beta}_{FD}=\left(\Delta X' P_{\Delta Z}\Delta X\right)^{-1}\Delta X' P_{\Delta Z}\Delta Y,$$

where

$$P_{\Delta Z}=\Delta Z(\Delta Z'\Delta Z)^{-1}\Delta Z'.$$

Equivalently, in moment notation, define

$$g_{FD,i}(b)=\sum_{t=2}^{T}\Delta z_{it}(\Delta y_{it}-b\Delta x_{it}).$$

Then $\widehat{\beta}_{FD}$ solves the just-identified or optimally weighted sample moment problem depending on the dimension of z. With a generic positive definite weighting matrix W_{FD},

$$\widehat{\beta}_{FD}=\arg\min_b\left[\frac{1}{N}\sum_i g_{FD,i}(b)\right]' W_{FD}\left[\frac{1}{N}\sum_i g_{FD,i}(b)\right].$$

Consistency. At the true β,

$$\Delta y_i-\beta\Delta x_i=\Delta v_i.$$

Hence

$$\mathbb{E}[g_{FD,i}(\beta)]=\mathbb{E}\left[\sum_{t=2}^{T}\Delta z_{it}\Delta v_{it}\right].$$

Because the original strict exogeneity condition implies

$$\mathbb{E}[v_{it}\mid z_{i1},\ldots,z_{iT},\eta_i]=0$$

for every t, it follows that

$$\mathbb{E}[\Delta v_{it} \mid z_{i1}, \ldots, z_{iT}, \eta_i] = 0.$$

Therefore,

$$\mathbb{E}[\Delta z_{it} \Delta v_{it}] = 0,$$

and the moment condition is valid.

Asymptotic distribution. Let

$$G_{FD} = -\mathbb{E}\left[\frac{\partial g_{FD,i}(\beta)}{\partial b}\right] = \mathbb{E}\left[\sum_{t=2}^{T} \Delta z_{it} \Delta x_{it}\right].$$

Let

$$S_{FD} = \mathbb{V}\mathrm{ar}(g_{FD,i}(\beta)) = \mathbb{E}\left[g_{FD,i}(\beta) g_{FD,i}(\beta)'\right].$$

For a generic GMM weighting matrix W_{FD},

$$\sqrt{N}(\widehat{\beta}_{FD} - \beta) \xrightarrow{d} N\left(0, (G'_{FD} W_{FD} G_{FD})^{-1} G'_{FD} W_{FD} S_{FD} W_{FD} G_{FD} (G'_{FD} W_{FD} G_{FD})^{-1}\right).$$

If the estimator is exactly identified, this simplifies to

$$\sqrt{N}(\widehat{\beta}_{FD} - \beta) \xrightarrow{d} N(0, G_{FD}^{-1} S_{FD} G_{FD}^{-1})$$

in the scalar moment case.

Within-group IV estimator

The within equation is

$$\widetilde{y}_i = \beta \widetilde{x}_i + \widetilde{v}_i,$$

where

$$\widetilde{y}_{it} = y_{it} - \bar{y}_i, \qquad \widetilde{x}_{it} = x_{it} - \bar{x}_i, \qquad \widetilde{z}_{it} = z_{it} - \bar{z}_i.$$

The within IV moment is

$$\mathbb{E}\left[\sum_{t=1}^{T} \widetilde{z}_{it}(\widetilde{y}_{it} - \beta \widetilde{x}_{it})\right] = 0.$$

The corresponding sample moment is

$$g_{WG,i}(b) = \sum_{t=1}^{T} \widetilde{z}_{it}(\widetilde{y}_{it} - b \widetilde{x}_{it}).$$

The 2SLS estimator is

$$\widehat{\beta}_{WG} = (\widetilde{X}' P_{\widetilde{Z}} \widetilde{X})^{-1} \widetilde{X}' P_{\widetilde{Z}} \widetilde{Y},$$

where $\widetilde{X}$, $\widetilde{Y}$, and $\widetilde{Z}$ are the stacked within-transformed variables.

Consistency. At the true β,

$$\widetilde{y}_i - \beta \widetilde{x}_i = \widetilde{v}_i.$$

Strict exogeneity gives

$$\mathbb{E}[\widetilde{v}_{it} \mid z_{i1}, \ldots, z_{iT}, \eta_i] = 0,$$

and therefore

$$\mathbb{E}[\widetilde{z}_{it} \widetilde{v}_{it}] = 0.$$

Thus the within IV moments are valid.

Asymptotic distribution. Define

$$G_{WG} = \mathbb{E}\left[\sum_{t=1}^{T} \widetilde{z}_{it} \widetilde{x}_{it}\right],$$

and

$$S_{WG} = \mathbb{V}\mathrm{ar}(g_{WG,i}(\beta)).$$

Then, for a generic weighting matrix W_{WG},

$$\sqrt{N}(\widehat{\beta}_{WG} - \beta) \xrightarrow{d} N\left(0, (G'_{WG} W_{WG} G_{WG})^{-1} G'_{WG} W_{WG} S_{WG} W_{WG} G_{WG} (G'_{WG} W_{WG} G_{WG})^{-1}\right).$$

With the efficient GMM weighting matrix $W_{WG} = S_{WG}^{-1}$, this becomes

$$\sqrt{N}(\widehat{\beta}_{WG} - \beta) \xrightarrow{d} N\left(0, (G'_{WG} S_{WG}^{-1} G_{WG})^{-1}\right).$$

1.2 Part (b): Optimal IV estimator under a linear first stage

Now assume the linear first-stage restriction

$$\mathbb{E}[x_{it} \mid z_{i1}, \ldots, z_{iT}] = \Pi' z_{it},$$

where Π is $r \times 1$ and full rank. In stacked form,

$$\mathbb{E}[x_i \mid z_i] = Z_i \Pi,$$

where Z_i is the $T \times r$ matrix whose t-th row is z'_{it}.

After removing fixed effects using some transformation A satisfying $A\iota_T = 0$, we have

$$Ay_i = \beta A x_i + A v_i.$$

The transformed error covariance is

$$\mathbb{V}\mathrm{ar}(Av_i \mid z_i) = A\Sigma(z_i)A'.$$

Optimal IV principle. For a linear model with conditional moment restrictions and heteroskedastic/serially correlated errors, the optimal instruments are proportional to the conditional expectation of the transformed endogenous regressor multiplied by the inverse conditional covariance matrix of the transformed error.

Thus, for a transformation A, the optimal instrument is

$$h_i^*(z_i) = \mathbb{E}[Ax_i \mid z_i]'[A\Sigma(z_i)A']^{-1}.$$

Since

$$\mathbb{E}[Ax_i \mid z_i] = AZ_i\Pi,$$

the optimal instrument becomes

$$h_i^*(z_i) = (AZ_i\Pi)'[A\Sigma(z_i)A']^{-1}.$$

Choice of transformation. Because the fixed effect is removed by any full-rank transformation A such that $A\iota_T = 0$, a convenient choice is the within transformation M_T, but M_T is singular. Alternatively, one may work in the $(T-1)$-dimensional subspace orthogonal to ι_T. Let Q be a $(T-1) \times T$ matrix such that

$$Q\iota_T = 0, \qquad QQ' = I_{T-1}.$$

Then the transformed model is

$$Qy_i = \beta Qx_i + Qv_i.$$

The conditional covariance matrix is

$$\Omega_i = Q\Sigma(z_i)Q'.$$

The optimal IV estimator can be written as GLS-IV:

$$\widehat{\beta}_{IV}^{opt} = \left[\sum_{i=1}^{N} \widehat{x}_i'\Omega_i^{-1}\widehat{x}_i\right]^{-1} \left[\sum_{i=1}^{N} \widehat{x}_i'\Omega_i^{-1}Qy_i\right],$$

where

$$\widehat{x}_i = \mathbb{E}[Qx_i \mid z_i] = QZ_i\Pi.$$

Therefore,

$$\boxed{\widehat{\beta}_{IV}^{opt} = \left[\sum_{i=1}^{N} (QZ_i\Pi)'\Omega_i^{-1}(Qx_i)\right]^{-1} \left[\sum_{i=1}^{N} (QZ_i\Pi)'\Omega_i^{-1}(Qy_i)\right]}$$

or, replacing Qx_i with its fitted value in the efficient population expression,

$$\boxed{\widehat{\beta}_{IV}^{opt} = \left[\sum_{i=1}^{N}(QZ_i\Pi)'\Omega_i^{-1}(QZ_i\Pi)\right]^{-1}\left[\sum_{i=1}^{N}(QZ_i\Pi)'\Omega_i^{-1}Qy_i\right].}$$

This is unfeasible because it depends on Π, ρ, and σ^2, which are unknown in practice.

Asymptotic variance. At the truth,

$$Qy_i = \beta Qx_i + Qv_i.$$

The optimal IV estimator has asymptotic variance

$$\boxed{V_{opt} = \left\{\mathbb{E}\left[(QZ_i\Pi)'\Omega_i^{-1}(QZ_i\Pi)\right]\right\}^{-1}.}$$

This formula follows from the fact that the optimal instrument exactly accounts for both the first-stage projection and the covariance structure of the transformed error.

Constructing Σ

The AR(1)-type process is

$$v_{it} = \rho v_{i,t-1} + \varepsilon_{it}, \qquad \mathbb{V}\mathrm{ar}(v_{i1}) = \sigma^2, \qquad \mathbb{V}\mathrm{ar}(\varepsilon_{it}) = \sigma^2.$$

For $t \geq 2$,

$$v_{it} = \rho^{t-1}v_{i1} + \sum_{s=2}^{t}\rho^{t-s}\varepsilon_{is}.$$

Therefore, for $t \geq s$,

$$\mathbb{C}\mathrm{ov}(v_{it}, v_{is}) = \rho^{t-s}\mathbb{V}\mathrm{ar}(v_{is}).$$

The variance satisfies

$$\mathbb{V}\mathrm{ar}(v_{it}) = \rho^2\mathbb{V}\mathrm{ar}(v_{i,t-1}) + \sigma^2.$$

Recursively,

$$\mathbb{V}\mathrm{ar}(v_{it}) = \sigma^2\sum_{\ell=0}^{t-1}\rho^{2\ell} = \sigma^2\frac{1-\rho^{2t}}{1-\rho^2}, \qquad \rho^2 \neq 1.$$

If $\rho = 1$, then

$$\mathbb{V}\mathrm{ar}(v_{it}) = t\sigma^2.$$

Thus, for $t \geq s$,

$$\Sigma_{ts} = \rho^{t-s}\sigma^2\sum_{\ell=0}^{s-1}\rho^{2\ell}.$$

By symmetry,

$$\Sigma_{ts} = \Sigma_{st}.$$

When are FD or WG efficient?

The FD estimator is efficient if the FD transformation and the instrument Δz_{it} happen to coincide with the optimal transformed instrument and weighting. This occurs in special cases where the transformed error covariance is proportional to the identity and the first-stage direction is aligned with the chosen instruments.

Similarly, WG is efficient if the within transformation and the within instruments span the same optimal instrument space and if the induced covariance matrix after within transformation is spherical up to scale.

The key cases are:

(i) If the transformed errors under FD are serially uncorrelated and homoskedastic, then FD can be efficient.

(ii) If the transformed errors under WG are serially uncorrelated and homoskedastic in the within space, then WG can be efficient.

(iii) If the original v_{it} are i.i.d. over time, within-group transformations are typically closer to efficient because first differencing creates moving-average serial correlation:

$$\Delta v_{it} = v_{it} - v_{i,t-1}.$$

Even if v_{it} is serially uncorrelated, Δv_{it} is negatively serially correlated.

(iv) If v_{it} is close to a random walk, then first differences remove much of the persistence and may be more efficient.

Thus:

$$\boxed{\text{WG tends to be relatively efficient when errors are weakly serially correlated,}}$$

while

$$\boxed{\text{FD tends to be relatively efficient when errors are highly persistent.}}$$

Neither FD nor WG is generally efficient unless the covariance structure happens to make their transformation and weighting coincide with the optimal GLS-IV transformation.

1.3 Part (c): Monte Carlo simulation

The problem asks for a Monte Carlo simulation, but this solution describes the algorithm rather than reporting numerical output.

The simulation design is:

$$n_{MC} = 10000, \qquad N = 2000, \qquad T = 7, \qquad \beta = 1.$$

Set

$$r = 2, \qquad \Pi = (1, 1)', \qquad \sigma^2 = 1, \qquad \alpha = 0.4.$$

For each $\rho \in \{-0.2, 0, 0.2, \ldots, 1\}$, repeat the following steps.

Step 1: Draw instruments. Draw

$$z_{it} \in \mathbb{R}^2$$

from a normal distribution. For example,

$$z_{it} \sim N(0, I_2).$$

The problem allows the researcher to choose the mean and variance.

Step 2: Generate errors. Initialize

$$v_{i1} \sim N(0, \sigma^2).$$

For $t = 2, \ldots, T$, draw

$$\varepsilon_{it} \sim N(0, \sigma^2),$$

and set

$$v_{it} = \rho v_{i,t-1} + \varepsilon_{it}.$$

Step 3: Generate endogenous regressor. Draw

$$u_{it} \sim N(0, \sigma^2),$$

independent of v_i and z_i. Generate

$$x_{it} = \Pi' z_{it} + \alpha \varepsilon_{it} + \sqrt{1 - \alpha^2} u_{it}.$$

The term $\alpha \varepsilon_{it}$ generates endogeneity because x_{it} is correlated with innovations to v_{it}. The instruments remain valid because z_{it} is drawn independently of the structural shocks.

Step 4: Generate outcome. Draw an individual effect η_i, possibly correlated with z_i if desired. Since both FD and WG remove η_i, its precise distribution does not affect consistency. Generate

$$y_{it} = \beta x_{it} + \eta_i + v_{it}.$$

Step 5: Estimate. For each simulated sample, compute:

$$\widehat{\beta}_{FD}, \qquad \widehat{\beta}_{WG}, \qquad \widehat{\beta}_{IV}^{opt}.$$

Step 6: Compute Monte Carlo variances. For each estimator and each ρ, compute

$$\widehat{\mathbb{V}\mathrm{ar}}_{MC}(\widehat{\beta}) = \frac{1}{n_{MC}-1} \sum_{m=1}^{n_{MC}} \left(\widehat{\beta}^{(m)} - \bar{\widehat{\beta}} \right)^2 .$$

The figure should plot

$$N\widehat{\mathbb{V}\mathrm{ar}}_{MC}(\widehat{\beta})$$

against ρ, separately for FD, WG, and optimal IV.

Expected qualitative result. The optimal IV estimator should have the lowest variance for each ρ, because it uses the correct covariance structure and the correct first-stage projection. FD and WG generally trade places depending on ρ. When ρ is high, FD is expected to become relatively more attractive. When ρ is low or close to zero, WG is expected to perform better.

1.4 Part (d): Main lesson

The main lesson is that eliminating fixed effects is not only about consistency. Both FD and WG remove η_i, and both can be consistent under strict exogeneity of the instruments. However, they are generally not equally efficient. Their relative efficiency depends on the serial correlation structure of the errors.

First differencing transforms errors into

$$\Delta v_{it} = v_{it} - v_{i,t-1}.$$

If v_{it} is close to a random walk, then Δv_{it} is close to the innovation and has relatively weak serial dependence. In that case, FD can be attractive.

Within transformation subtracts individual means:

$$v_{it} - \bar{v}_i.$$

This transformation is often attractive when v_{it} is weakly serially correlated, because it uses deviations from individual means without inducing as much strong moving-average structure as differencing does.

Therefore, in empirical work:

$$\boxed{\text{Choose FD if errors are believed to be highly persistent.}}$$

Choose WG if:

$$\boxed{\text{Errors are believed to be weakly serially correlated or approximately i.i.d. over time.}}$$

The optimal IV estimator dominates both when the covariance structure and first stage are correctly specified, but it is generally unfeasible and more model-dependent.

2 Exercise 2: Income and Consumption

The data contain panel observations on income, consumption, wealth, and age for $N = 792$ households over $T = 6$ biennial waves. The model decomposes income into permanent and transitory components:

$$y_{it} = \eta_{it} + \varepsilon_{it}.$$

The permanent component follows a random walk:

$$\eta_{it} = \eta_{i,t-1} + u_{it}.$$

The transitory component follows an MA(1):

$$\varepsilon_{it} = v_{it} + \psi_\varepsilon v_{i,t-1}.$$

The shocks u_{it} and v_{it} are serially uncorrelated, mutually orthogonal at all leads and lags, and have constant variances:

$$\mathbb{V}\text{ar}(u_{it}) = \sigma_\eta^2, \qquad \mathbb{V}\text{ar}(v_{it}) = \sigma_\varepsilon^2.$$

Consumption growth is modeled as

$$\Delta c_{it} = \alpha u_{it} + \alpha\beta v_{it} + (e_{it} + \psi_c e_{i,t-1}),$$

where

$$\mathbb{V}\text{ar}(e_{it}) = \sigma_c^2,$$

and e_{it} is orthogonal to u_{it} and v_{it} at all leads and lags.

The parameter vector is

$$\theta = \left(\sigma_\eta^2, \psi_\varepsilon, \sigma_\varepsilon^2, \alpha, \beta, \psi_c, \sigma_c\right).$$

The vector of observed growth rates is

$$s_i = (\Delta y_{i2}, \Delta c_{i2}, \ldots, \Delta y_{iT}, \Delta c_{iT})'.$$

Since $T = 6$, there are $T - 1 = 5$ income growth observations and 5 consumption growth observations, so s_i is 10×1.

2.1 Part (a): Derive $\Omega(\theta) = \mathbb{V}\text{ar}(s_i)$

We first derive the covariance structure of income growth, consumption growth, and their cross-covariances.

Income growth

Since

$$y_{it} = \eta_{it} + \varepsilon_{it},$$

we have

$$\Delta y_{it} = y_{it} - y_{i,t-1} = \Delta\eta_{it} + \Delta\varepsilon_{it}.$$

The permanent component satisfies

$$\Delta\eta_{it} = u_{it}.$$

The transitory component is

$$\varepsilon_{it} = v_{it} + \psi_\varepsilon v_{i,t-1}.$$

Thus

$$\Delta\varepsilon_{it} = \varepsilon_{it} - \varepsilon_{i,t-1} = (v_{it} + \psi_\varepsilon v_{i,t-1}) - (v_{i,t-1} + \psi_\varepsilon v_{i,t-2}).$$

Therefore,

$$\Delta\varepsilon_{it} = v_{it} + (\psi_\varepsilon - 1)v_{i,t-1} - \psi_\varepsilon v_{i,t-2}.$$

Hence

$$\boxed{\Delta y_{it} = u_{it} + v_{it} + (\psi_\varepsilon - 1)v_{i,t-1} - \psi_\varepsilon v_{i,t-2}.}$$

Let

$$A_t = v_{it} + (\psi_\varepsilon - 1)v_{i,t-1} - \psi_\varepsilon v_{i,t-2}.$$

Then

$$\Delta y_{it} = u_{it} + A_t.$$

Because u and v are mutually orthogonal,

$$\mathbb{C}\mathrm{ov}(\Delta y_{it}, \Delta y_{is}) = \mathbb{C}\mathrm{ov}(u_{it}, u_{is}) + \mathbb{C}\mathrm{ov}(A_t, A_s).$$

Since u_{it} is serially uncorrelated,

$$\mathbb{C}\mathrm{ov}(u_{it}, u_{is}) = \begin{cases} \sigma_\eta^2, & t = s, \\ 0, & t \neq s. \end{cases}$$

Now compute the covariance of A_t. The coefficients of A_t on $v_{it}, v_{i,t-1}, v_{i,t-2}$ are

$$1, \quad \psi_\varepsilon - 1, \quad -\psi_\varepsilon.$$

Variance.

$$\begin{aligned}\mathbb{V}\mathrm{ar}(A_t) &= \sigma_\varepsilon^2\left[1^2 + (\psi_\varepsilon - 1)^2 + \psi_\varepsilon^2\right] \\ &= \sigma_\varepsilon^2\left[1 + (\psi_\varepsilon^2 - 2\psi_\varepsilon + 1) + \psi_\varepsilon^2\right] \\ &= \sigma_\varepsilon^2\left[2 + 2\psi_\varepsilon^2 - 2\psi_\varepsilon\right].\end{aligned}$$

Thus

$$\boxed{\mathbb{V}\mathrm{ar}(\Delta y_{it}) = \sigma_\eta^2 + \sigma_\varepsilon^2\left[2 + 2\psi_\varepsilon^2 - 2\psi_\varepsilon\right].}$$

Lag-one covariance. We compute

$$\mathbb{C}\mathrm{ov}(A_t, A_{t-1}).$$

Now

$$A_{t-1} = v_{i,t-1} + (\psi_\varepsilon - 1)v_{i,t-2} - \psi_\varepsilon v_{i,t-3}.$$

The overlapping shocks are $v_{i,t-1}$ and $v_{i,t-2}$. Therefore,

$$\begin{aligned}\mathbb{C}\mathrm{ov}(A_t, A_{t-1}) &= \sigma_\varepsilon^2\left[(\psi_\varepsilon - 1)\cdot 1 + (-\psi_\varepsilon)(\psi_\varepsilon - 1)\right] \\ &= \sigma_\varepsilon^2(\psi_\varepsilon - 1)(1 - \psi_\varepsilon) \\ &= -\sigma_\varepsilon^2(1 - \psi_\varepsilon)^2.\end{aligned}$$

Thus

$$\boxed{\mathbb{C}\mathrm{ov}(\Delta y_{it}, \Delta y_{i,t-1}) = -\sigma_\varepsilon^2(1 - \psi_\varepsilon)^2.}$$

Lag-two covariance. Now compare

$$A_t = v_t + (\psi_\varepsilon - 1)v_{t-1} - \psi_\varepsilon v_{t-2}$$

and

$$A_{t-2} = v_{t-2} + (\psi_\varepsilon - 1)v_{t-3} - \psi_\varepsilon v_{t-4}.$$

The only overlap is v_{t-2}. Hence

$$\mathbb{C}\mathrm{ov}(A_t, A_{t-2}) = (-\psi_\varepsilon)(1)\sigma_\varepsilon^2.$$

Therefore

$$\boxed{\mathbb{C}\mathrm{ov}(\Delta y_{it}, \Delta y_{i,t-2}) = -\psi_\varepsilon\sigma_\varepsilon^2.}$$

Higher lags. For $|t - s| > 2$, there are no overlapping v-shocks, so

$$\boxed{\mathbb{C}\mathrm{ov}(\Delta y_{it}, \Delta y_{is}) = 0 \qquad \text{if } |t - s| > 2.}$$

Consumption growth

Consumption growth is

$$\Delta c_{it} = \alpha u_{it} + \alpha\beta v_{it} + e_{it} + \psi_c e_{i,t-1}.$$

Let

$$B_t = \alpha u_t + \alpha\beta v_t + e_t + \psi_c e_{t-1}.$$

Then

$$\Delta c_t = B_t.$$

Variance. Because u_t, v_t, e_t, e_{t-1} are mutually orthogonal,

$$\begin{aligned}\mathbb{V}\mathrm{ar}(\Delta c_t) &= \alpha^2\sigma_\eta^2 + \alpha^2\beta^2\sigma_\varepsilon^2 + \sigma_c^2 + \psi_c^2\sigma_c^2 \\ &= \boxed{\alpha^2\sigma_\eta^2 + \alpha^2\beta^2\sigma_\varepsilon^2 + (1+\psi_c^2)\sigma_c^2.}\end{aligned}$$

Lag-one covariance. The only overlap between Δc_t and Δc_{t-1} is e_{t-1}. Specifically,

$$\Delta c_{t-1} = \alpha u_{t-1} + \alpha\beta v_{t-1} + e_{t-1} + \psi_c e_{t-2}.$$

Thus

$$\boxed{\mathbb{C}\mathrm{ov}(\Delta c_t, \Delta c_{t-1}) = \psi_c\sigma_c^2.}$$

For lags greater than one,

$$\boxed{\mathbb{C}\mathrm{ov}(\Delta c_t, \Delta c_s) = 0 \qquad \text{if } |t-s| > 1.}$$

Income-consumption covariance

We compute

$$\mathbb{C}\mathrm{ov}(\Delta y_t, \Delta c_s).$$

Recall

$$\Delta y_t = u_t + v_t + (\psi_\varepsilon - 1)v_{t-1} - \psi_\varepsilon v_{t-2},$$

and

$$\Delta c_s = \alpha u_s + \alpha\beta v_s + e_s + \psi_c e_{s-1}.$$

Since e is orthogonal to income shocks, it does not enter the covariance.

Same period. For $s = t$,

$$\begin{aligned}\mathbb{C}\mathrm{ov}(\Delta y_t, \Delta c_t) &= \mathbb{C}\mathrm{ov}(u_t, \alpha u_t) + \mathbb{C}\mathrm{ov}(v_t, \alpha\beta v_t) \\ &= \alpha\sigma_\eta^2 + \alpha\beta\sigma_\varepsilon^2.\end{aligned}$$

Thus

$$\boxed{\mathbb{C}\mathrm{ov}(\Delta y_t, \Delta c_t) = \alpha\sigma_\eta^2 + \alpha\beta\sigma_\varepsilon^2.}$$

Consumption one period earlier. For $s = t - 1$,

$$\Delta c_{t-1} = \alpha u_{t-1} + \alpha\beta v_{t-1} + e_{t-1} + \psi_c e_{t-2}.$$

The only overlap with Δy_t is v_{t-1}. Hence

$$\boxed{\mathbb{C}\mathrm{ov}(\Delta y_t, \Delta c_{t-1}) = \alpha\beta(\psi_\varepsilon - 1)\sigma_\varepsilon^2.}$$

Consumption two periods earlier. For $s = t - 2$, the overlap is v_{t-2}, so

$$\boxed{\mathbb{C}\mathrm{ov}(\Delta y_t, \Delta c_{t-2}) = -\alpha\beta\psi_\varepsilon\sigma_\varepsilon^2.}$$

Consumption one period later. For $s = t + 1$, Δc_{t+1} contains u_{t+1} and v_{t+1}, which do not overlap with Δy_t. Therefore,

$$\boxed{\mathbb{C}\mathrm{ov}(\Delta y_t, \Delta c_{t+1}) = 0.}$$

Likewise,

$$\mathbb{C}\mathrm{ov}(\Delta y_t, \Delta c_s) = 0$$

for all other non-overlapping pairs.

Matrix representation

Let

$$m = T - 1.$$

Define

$$\Delta y_i = (\Delta y_{i2}, \ldots, \Delta y_{iT})', \qquad \Delta c_i = (\Delta c_{i2}, \ldots, \Delta c_{iT})'.$$

Then

$$s_i = \begin{pmatrix} \Delta y_{i2} \\ \Delta c_{i2} \\ \vdots \\ \Delta y_{iT} \\ \Delta c_{iT} \end{pmatrix}.$$

Equivalently, one may stack all income growth observations first and then all consumption growth observations:

$$\tilde{s}_i = \begin{pmatrix} \Delta y_i \\ \Delta c_i \end{pmatrix}.$$

Then

$$\mathbb{V}\mathrm{ar}(\tilde{s}_i) = \begin{pmatrix} \Omega_{yy} & \Omega_{yc} \\ \Omega_{cy} & \Omega_{cc} \end{pmatrix}.$$

The entries are:

$$(\Omega_{yy})_{ts} = \begin{cases} \sigma_\eta^2 + \sigma_\varepsilon^2(2 + 2\psi_\varepsilon^2 - 2\psi_\varepsilon), & t = s, \\ -\sigma_\varepsilon^2(1 - \psi_\varepsilon)^2, & |t - s| = 1, \\ -\psi_\varepsilon \sigma_\varepsilon^2, & |t - s| = 2, \\ 0, & |t - s| > 2. \end{cases}$$

Here $t, s \in \{2, \ldots, T\}$.

$$(\Omega_{cc})_{ts} = \begin{cases} \alpha^2\sigma_\eta^2 + \alpha^2\beta^2\sigma_\varepsilon^2 + (1 + \psi_c^2)\sigma_c^2, & t = s, \\ \psi_c \sigma_c^2, & |t - s| = 1, \\ 0, & |t - s| > 1. \end{cases}$$

The income-consumption block has entries

$$(\Omega_{yc})_{ts} = \mathbb{C}\mathrm{ov}(\Delta y_t, \Delta c_s).$$

Using the results above,

$$(\Omega_{yc})_{ts} = \begin{cases} \alpha\sigma_\eta^2 + \alpha\beta\sigma_\varepsilon^2, & s = t, \\ \alpha\beta(\psi_\varepsilon - 1)\sigma_\varepsilon^2, & s = t - 1, \\ -\alpha\beta\psi_\varepsilon\sigma_\varepsilon^2, & s = t - 2, \\ 0, & \text{otherwise}. \end{cases}$$

Finally,

$$\Omega_{cy} = \Omega_{yc}'.$$

2.2 Part (b): Estimation through covariance matching

The model maps the structural parameter vector θ into a covariance matrix:

$$\theta \mapsto \Omega(\theta).$$

The empirical analogue is the sample covariance matrix of s_i:

$$\widehat{\Omega} = \frac{1}{N}\sum_{i=1}^{N}(s_i - \bar{s})(s_i - \bar{s})'.$$

The idea is to choose θ so that the model-implied covariance matrix matches the sample covariance matrix as closely as possible.

Let

$$\widehat{m} = \text{vech}(\widehat{\Omega})$$

be the vector of distinct elements of the sample covariance matrix, and let

$$m(\theta) = \text{vech}(\Omega(\theta)).$$

A minimum-distance estimator solves

$$\widehat{\theta} = \arg\min_{\theta} \left[\widehat{m} - m(\theta)\right]' W_N \left[\widehat{m} - m(\theta)\right],$$

where W_N is a positive definite weighting matrix.

Identity weighting. A simple estimator uses $W_N = I$. This is easy to implement and often stable.

Efficient weighting. An asymptotically efficient estimator uses

$$W_N \approx \mathbb{V}\text{ar}(\sqrt{N}\widehat{m})^{-1}.$$

This can be estimated using the empirical variance of the outer-product terms.

Standard errors. Let

$$D(\theta) = \frac{\partial m(\theta)}{\partial \theta'}.$$

Under standard minimum-distance theory,

$$\sqrt{N}(\widehat{\theta} - \theta_0) \xrightarrow{d} N(0, V_\theta),$$

where

$$V_\theta = (D'WD)^{-1} D'WSWD(D'WD)^{-1}.$$

Here S is the asymptotic variance of $\sqrt{N}(\widehat{m} - m(\theta_0))$. If $W = S^{-1}$, then

$$V_\theta = (D'S^{-1}D)^{-1}.$$

2.3 Part (c): Estimating the income process only

Using only income data means we use the covariance matrix of

$$\Delta y_i = (\Delta y_{i2}, \ldots, \Delta y_{iT})'.$$

The parameters are

$$\theta_y = (\sigma_\eta^2, \psi_\varepsilon, \sigma_\varepsilon^2).$$

The model-implied covariance matrix $\Omega_{yy}(\theta_y)$ is described above.

The minimum-distance estimator is

$$\widehat{\theta}_y = \arg\min_{\theta_y} \left[\text{vech}(\widehat{\Omega}_{yy}) - \text{vech}(\Omega_{yy}(\theta_y))\right]' W_y \left[\text{vech}(\widehat{\Omega}_{yy}) - \text{vech}(\Omega_{yy}(\theta_y))\right].$$

Interpretation of $\sigma_\eta^2/\sigma_\varepsilon^2$. The ratio

$$\frac{\sigma_\eta^2}{\sigma_\varepsilon^2}$$

measures the relative importance of permanent income shocks compared with transitory shocks. If this ratio is large, most income growth variation comes from persistent innovations u_{it}. If it is small, income growth is mostly driven by transitory shocks v_{it}.

Heuristic model fit. To compare the estimated covariance matrix and the data covariance matrix, one should place side by side:

$$\widehat{\Omega}_{yy} \qquad \text{and} \qquad \Omega_{yy}(\widehat{\theta}_y).$$

The model has a very specific covariance structure:

- diagonal elements are constant;
- first off-diagonal elements are constant;
- second off-diagonal elements are constant;
- all further off-diagonal elements are zero.

If the empirical covariance matrix exhibits meaningful nonzero covariance beyond lag two, then the transitory component may require richer dynamics than MA(1). For example, an MA(2) or ARMA process may improve fit.

If the diagonal elements vary across time, then the assumption of constant variances may be too restrictive. Time-varying variances or age-dependent shock variances could improve fit.

If income growth covariances display long-run positive dependence, the permanent component may need richer dynamics than a pure random walk. For example, the persistent component could follow a more general ARIMA process.

2.4 Part (d): Full covariance estimation and interpretation

The full parameter vector is

$$\theta = \left(\sigma_\eta^2, \psi_\varepsilon, \sigma_\varepsilon^2, \alpha, \beta, \psi_c, \sigma_c\right).$$

Using the full covariance structure means matching the covariance matrix of both income growth and consumption growth:

$$\Omega(\theta) = \begin{pmatrix} \Omega_{yy}(\theta) & \Omega_{yc}(\theta) \\ \Omega_{cy}(\theta) & \Omega_{cc}(\theta) \end{pmatrix}.$$

The estimator solves

$$\widehat{\theta} = \arg\min_{\theta} \left[\mathrm{vech}(\widehat{\Omega}) - \mathrm{vech}(\Omega(\theta))\right]' W \left[\mathrm{vech}(\widehat{\Omega}) - \mathrm{vech}(\Omega(\theta))\right].$$

Interpretation of α. The parameter α measures the response of consumption growth to permanent income shocks:

$$\Delta c_{it} = \alpha u_{it} + \cdots .$$

Under the permanent income hypothesis, consumption should respond strongly to permanent shocks because permanent shocks change lifetime resources. Therefore, one expects

$$\alpha \approx 1$$

if the permanent income hypothesis holds exactly and if income is measured in the same units as consumption.

Interpretation of β. The response to transitory shocks is

$$\alpha\beta v_{it}.$$

If $\alpha \approx 1$, then β is approximately the marginal propensity to consume out of a transitory shock relative to a permanent shock. The permanent income hypothesis predicts that consumption should respond less to transitory shocks than to permanent shocks. Hence one expects

$$0 \leq \beta < 1.$$

A small β indicates strong consumption smoothing against transitory income shocks. A large β suggests excess sensitivity of consumption to transitory income fluctuations, possibly due to borrowing constraints, liquidity constraints, precautionary saving motives, or measurement error.

Conclusion about the permanent income hypothesis. If estimates show:

$$\widehat{\alpha} \text{ close to } 1, \qquad \widehat{\beta} \text{ close to } 0,$$

then the evidence is broadly consistent with the permanent income hypothesis.

If instead:

$$\widehat{\beta} > 0$$

and economically large, then consumption responds to transitory income shocks, which is inconsistent with the simplest version of the permanent income hypothesis.

3 Exercise 3: Testing for Correlated Random Coefficients

We consider

$$y_{it} = \alpha_i + \beta_i x_{it} + \gamma' z_{it} + v_{it},$$

with strict exogeneity:

$$\mathbb{E}[v_{it} \mid \beta_i, x_{i1}, z_{i1}, \ldots, x_{iT}, z_{iT}] = 0.$$

Let

$$w_{it} = \begin{pmatrix} x_{it} \\ z_{it} \end{pmatrix}, \qquad \delta_i = \begin{pmatrix} \beta_i \\ \gamma \end{pmatrix}.$$

The slope on x_{it} is heterogeneous, whereas the slope on z_{it} is homogeneous.

3.1 Part (a): Probability limit of the pooled first-difference estimator

First difference the model:

$$\Delta y_{it} = \beta_i \Delta x_{it} + \gamma' \Delta z_{it} + \Delta v_{it}.$$

Equivalently,

$$\Delta y_{it} = \Delta w_{it}' \begin{pmatrix} \beta_i \\ \gamma \end{pmatrix} + \Delta v_{it}.$$

The researcher incorrectly estimates a common-slope first-difference regression:

$$\widehat{\delta} = \begin{pmatrix} \widehat{\beta} \\ \widehat{\gamma} \end{pmatrix} = \left(\sum_{i=1}^{N} \sum_{t=1}^{T} \Delta w_{it} \Delta w_{it}' \right)^{-1} \left(\sum_{i=1}^{N} \sum_{t=1}^{T} \Delta w_{it} \Delta y_{it} \right).$$

Strictly speaking, first differences are defined for $t = 2, \ldots, T$, but the notation in the question sums over t. We interpret Δw_{it} and Δy_{it} as available differenced observations.

Substitute the true model:

$$\Delta y_{it} = \Delta x_{it} \beta_i + \Delta z_{it}' \gamma + \Delta v_{it}.$$

Then

$$\Delta w_{it} \Delta y_{it} = \Delta w_{it} \Delta x_{it} \beta_i + \Delta w_{it} \Delta z_{it}' \gamma + \Delta w_{it} \Delta v_{it}.$$

But

$$\Delta w_{it} \Delta z_{it}' \gamma$$

is included in

$$\Delta w_{it} \Delta w_{it}' \begin{pmatrix} 0 \\ \gamma \end{pmatrix}.$$

More directly, write

$$\Delta y_{it} = \Delta w_{it}' \begin{pmatrix} \bar{\beta} \\ \gamma \end{pmatrix} + \Delta x_{it} (\beta_i - \bar{\beta}) + \Delta v_{it},$$

where

$$\bar{\beta} = \mathbb{E}[\beta_i].$$

Therefore,

$$\widehat{\delta} = \begin{pmatrix} \bar{\beta} \\ \gamma \end{pmatrix} + \left(\sum_{i,t} \Delta w_{it} \Delta w_{it}' \right)^{-1} \sum_{i,t} \Delta w_{it} \Delta x_{it} (\beta_i - \bar{\beta}) + \left(\sum_{i,t} \Delta w_{it} \Delta w_{it}' \right)^{-1} \sum_{i,t} \Delta w_{it} \Delta v_{it}.$$

By strict exogeneity,

$$\mathbb{E}[\Delta w_{it} \Delta v_{it}] = 0.$$

Thus, as $N \to \infty$ with T fixed,

$$\text{plim} \widehat{\delta} = \begin{pmatrix} \bar{\beta} \\ \gamma \end{pmatrix} + Q^{-1} q,$$

where

$$Q = \mathbb{E} \left[\sum_{t=2}^{T} \Delta w_{it} \Delta w_{it}' \right],$$

and

$$q = \mathbb{E} \left[\sum_{t=2}^{T} \Delta w_{it} \Delta x_{it} (\beta_i - \bar{\beta}) \right].$$

Equivalently,

$$\boxed{\text{plim} \begin{pmatrix} \widehat{\beta} \\ \widehat{\gamma} \end{pmatrix} = \begin{pmatrix} \bar{\beta} \\ \gamma \end{pmatrix} + Q^{-1} \mathbb{E} \left[\sum_{t=2}^{T} \Delta w_{it} \Delta x_{it} (\beta_i - \bar{\beta}) \right].}$$

This shows that the pooled first-difference regression generally does not estimate the average slope $\bar{\beta}$ or the structural coefficient γ. The asymptotic bias depends on the covariance between the random coefficient β_i and functions of the regressors.

If

$$\mathbb{E}[\beta_i \mid w_i] = \bar{\beta},$$

then

$$\begin{aligned} q &= \mathbb{E} \left[\sum_{t} \Delta w_{it} \Delta x_{it} \mathbb{E}[(\beta_i - \bar{\beta}) \mid w_i] \right] \\ &= 0. \end{aligned}$$

Hence,

$$\boxed{\text{plim} \widehat{\beta} = \bar{\beta}, \qquad \text{plim} \widehat{\gamma} = \gamma.}$$

Thus the random-effects-type assumption

$$\mathbb{E}[\beta_i \mid w_i] = \bar{\beta}$$

restores consistency of the pooled first-difference estimator for the average slope and the homogeneous coefficient.

3.2 Part (b): Deriving $g(w_i)$ and a test

From part (a), the asymptotic bias is

$$Q^{-1}\mathbb{E}\left[\sum_{t=2}^{T}\Delta w_{it}\Delta x_{it}(\beta_i-\bar{\beta})\right].$$

Therefore, consistency for $(\bar{\beta},\gamma)$ is equivalent to

$$\mathbb{E}\left[\sum_{t=2}^{T}\Delta w_{it}\Delta x_{it}(\beta_i-\bar{\beta})\right]=0.$$

This can be written as

$$\mathbb{C}\mathrm{ov}(\beta_i,g(w_i))=0,$$

where

$$\boxed{g(w_i)=\sum_{t=2}^{T}\Delta w_{it}\Delta x_{it}.}$$

Here $g(w_i)$ is a $(k+1)\times 1$ vector because Δw_{it} is $(k+1)\times 1$ and Δx_{it} is scalar.

Indeed,

$$\mathbb{C}\mathrm{ov}(\beta_i,g(w_i))=\mathbb{E}[(\beta_i-\bar{\beta})g(w_i)]$$

and

$$\mathbb{E}[(\beta_i-\bar{\beta})g(w_i)]=\mathbb{E}\left[\sum_{t=2}^{T}\Delta w_{it}\Delta x_{it}(\beta_i-\bar{\beta})\right].$$

Conditional expectation model

Now suppose

$$\mathbb{E}[\beta_i\mid w_i]=\lambda_0+\lambda_1' g(w_i),$$

where

$$\lambda_1\in\mathbb{R}^{k+1}.$$

The null hypothesis is

$$H_0:\lambda_1=0_{k+1}.$$

Under H_0,

$$\mathbb{E}[\beta_i\mid w_i]=\lambda_0.$$

Since

$$\mathbb{E}[\beta_i]=\bar{\beta},$$

we have $\lambda_0 = \bar{\beta}$. Thus H_0 is the random-effects-type restriction that the heterogeneous slope is mean independent of the relevant function of the regressors.

How to estimate the model and test H_0

The individual-specific model in first differences is

$$\Delta y_{it} = \beta_i \Delta x_{it} + \gamma' \Delta z_{it} + \Delta v_{it}.$$

The key difficulty is that β_i is not directly observed. However, because $T \geq 3$, each individual has at least two first-differenced observations. In principle, one can construct individual-level estimates of slopes or use moment restrictions implied by the model.

A practical testing approach is based on the probability-limit expression derived above.

The pooled FD estimator is consistent for $(\bar{\beta}, \gamma)$ under H_0. If H_0 fails, the pooled FD estimator converges to a weighted slope that differs from $(\bar{\beta}, \gamma)$. Therefore, one can test for correlation between random coefficients and regressors by augmenting the differenced regression with interactions between Δx_{it} and $g(w_i)$.

Substitute the conditional mean model:

$$\beta_i = \lambda_0 + \lambda_1' g(w_i) + \xi_i,$$

where

$$\mathbb{E}[\xi_i \mid w_i] = 0.$$

Then the differenced equation becomes

$$\begin{aligned} \Delta y_{it} &= \left[\lambda_0 + \lambda_1' g(w_i) + \xi_i\right] \Delta x_{it} + \gamma' \Delta z_{it} + \Delta v_{it} \\ &= \lambda_0 \Delta x_{it} + \gamma' \Delta z_{it} + \left[g(w_i) \Delta x_{it}\right]' \lambda_1 + \xi_i \Delta x_{it} + \Delta v_{it}. \end{aligned}$$

Since

$$\mathbb{E}[\xi_i \mid w_i] = 0$$

and

$$\mathbb{E}[\Delta v_{it} \mid w_i, \beta_i] = 0,$$

the composite error

$$\xi_i \Delta x_{it} + \Delta v_{it}$$

is orthogonal to the regressors in the augmented equation.

Thus, estimate the augmented first-difference regression

$$\boxed{\Delta y_{it} = \lambda_0 \Delta x_{it} + \gamma' \Delta z_{it} + \left[g(w_i) \Delta x_{it}\right]' \lambda_1 + \text{error}_{it}.}$$

Then test

$$H_0 : \lambda_1 = 0.$$

Implementation details.

1. Construct first differences Δy_{it}, Δx_{it}, and Δz_{it}.
2. For each individual, construct
$$g(w_i) = \sum_{t=2}^{T} \Delta w_{it} \Delta x_{it}.$$
3. For every observation (i, t), construct interactions
$$\Delta x_{it} g(w_i).$$
4. Run OLS of Δy_{it} on:
$$\Delta x_{it}, \quad \Delta z_{it}, \quad \Delta x_{it} g(w_i).$$
5. Cluster standard errors at the individual level, because observations are serially correlated within individual and because $g(w_i)$ is common across all observations for a given individual.
6. Conduct a Wald test of the joint null:
$$H_0 : \lambda_1 = 0_{k+1}.$$

Interpretation. If the null is rejected, there is evidence that the individual-specific slope β_i is correlated with the regressor history w_i through the function $g(w_i)$. In that case, the pooled FD estimator is not consistent for $\bar{\beta}$ and γ. If the null is not rejected, the pooled FD approach is supported by the data.

3.3 Part (c): Application to income, time trend, and age squared

The empirical application sets:

$$y_{it} = \text{income}_{it}, \qquad x_{it} = t, \qquad z_{it} = \text{age}_{it}^2.$$

The model is

$$income_{it} = \alpha_i + \beta_i t + \gamma age_{it}^2 + v_{it}.$$

Here β_i is the individual-specific income growth trend. The question is whether this individual-specific income trend is correlated with the history of age squared.

First difference the model:

$$\Delta income_{it} = \beta_i \Delta t + \gamma \Delta age_{it}^2 + \Delta v_{it}.$$

Since $x_{it} = t$, we have

$$\Delta x_{it} = 1.$$

Also,

$$w_{it} = \begin{pmatrix} t \\ age_{it}^2 \end{pmatrix}.$$

Therefore,

$$\Delta w_{it} = \begin{pmatrix} 1 \\ \Delta age_{it}^2 \end{pmatrix}.$$

The function $g(w_i)$ is

$$\begin{aligned} g(w_i) &= \sum_{t=2}^{T} \Delta w_{it} \Delta x_{it} \\ &= \sum_{t=2}^{T} \begin{pmatrix} 1 \\ \Delta age_{it}^2 \end{pmatrix} \cdot 1 \\ &= \begin{pmatrix} T-1 \\ age_{iT}^2 - age_{i1}^2 \end{pmatrix}. \end{aligned}$$

The first component $T-1$ is constant across individuals and is absorbed into the intercept/slope term. The relevant source of heterogeneity is

$$age_{iT}^2 - age_{i1}^2.$$

The augmented regression is

$$\Delta income_{it} = \lambda_0 + \gamma \Delta age_{it}^2 + \lambda_{1,1}(T-1) + \lambda_{1,2}(age_{iT}^2 - age_{i1}^2) + \text{error}_{it}.$$

Because $\Delta x_{it} = 1$, the interaction $\Delta x_{it} g(w_i)$ is just $g(w_i)$. Since $T-1$ is constant, the empirically meaningful test is whether the coefficient on

$$age_{iT}^2 - age_{i1}^2$$

is zero.

Test. The null hypothesis is

$$H_0 : \lambda_{1,2} = 0.$$

If one includes all components of $g(w_i)$, the formal null is

$$H_0 : \lambda_1 = 0.$$

Because one component is constant, the test effectively reduces to the coefficient on the nonconstant part.

Interpretation. If H_0 is rejected, then there is evidence that heterogeneous income trends are correlated with age-squared histories. In that case, the pooled first-difference estimator is not appropriate for estimating the mean slope. A more robust approach would allow correlated random coefficients or estimate individual trends and then average them.

If H_0 is not rejected, then the pooled first-difference estimator is supported, and one can report the pooled FD estimate of the mean income trend with cluster-robust standard errors.

Mean slope estimate. Under the null, the mean slope is estimated by the coefficient on Δt in the pooled FD regression:

$$\Delta income_{it} = \bar{\beta} + \gamma \Delta age_{it}^2 + \text{error}_{it}.$$

Since $\Delta t = 1$, the intercept in this differenced regression corresponds to the mean slope $\bar{\beta}$, after controlling for Δage^2.

If the null is rejected, a correlated-random-coefficients specification is more appropriate. In that case, the average slope can be obtained from the augmented model by averaging the conditional mean:

$$\mathbb{E}[\beta_i] = \lambda_0 + \lambda_1' \mathbb{E}[g(w_i)].$$

The standard error can be obtained using the delta method or by bootstrap clustered at the individual level.